# Psychology and Cognitive Science: What Happens When Cognition Shifts

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# Psychology and Cognitive Science: What Happens When Cognition Shifts

## 1. Purpose and Question of the Investigation

Figure 1: Psychology and Cognitive Science: What Happens When Cognition Shifts —

This report presents the results of an investigation into the following question, using major theoretical systems in psychology and cognitive science.

> Do the theories of this domain correspond structurally to the five-stage model, Field -> Wave -> Edge -> Vortex -> Bundle?

Psychology and cognitive science are disciplines that describe how human cognition is formed, transformed, and stabilized. Why does insight arrive suddenly? How are conflicting beliefs handled? Under what conditions does deep immersion arise? The purpose of this investigation is to examine how the structure of the five-stage model corresponds to existing theories addressing these questions.

This document is a report of findings. It is not a psychology textbook, nor is it a document intended to prove the five-stage model correct. Places where structural correspondence appears strong and places where it remains weak are reported with the same level of restraint.

## 2. Method of Investigation

### Selection Criteria

Ten theories were selected from psychology and cognitive science. The aim was to cover both cognition as framework, relatively static structure, and cognition as temporal development, dynamic process.

- **Theories foregrounding processes of cognitive transformation**: insight problem solving, cognitive dissonance, and flow experience
- **Theories describing the environmental conditions under which cognition becomes possible**: 4E cognition, affordances, and conceptual metaphor
- **Computational frameworks bridging the two**: the Bayesian brain hypothesis, schema theory, biased competition of attention, and cognitive flexibility

### Method of Evaluation

For each theory, structural correspondence was examined from multiple independent viewpoints and the strength of correspondence was judged on that basis. Three evaluation lenses were used.

1. **Discriminating power of Edge**: how opposing elements enter into relation within cognitive process
2. **Mode of error processing**: how mismatches between prediction and reality are handled
3. **Presence or absence of consciousness**: whether the process proceeds latently, is experienced as conscious discomfort, or is experienced as conscious immersion

### Criteria of Judgment

- **Strong structural correspondence**: Clear correspondence is confirmed between multiple stages of the five-stage model and stage structures internal to the theory itself.
- **Partial correspondence**: Some stages correspond clearly, but the rest remain at a more conceptual level.
- **Conditional correspondence**: The correspondence holds only under a particular interpretation of the theory.

### Limits of the Investigation

This investigation examines whether structural correspondence is present. It does not claim causal relations. Descriptions of the theories remain at the level of overview, and have not been subjected to specialist review in each field. Multiple-angle review was conducted, but expert human review remains a task for future work.

## 3. Overview of the Five-Stage Model

The five-stage model used as the reference in this investigation is defined as follows.

**Field**: An undifferentiated condition. It refers to the initial state in which neither direction nor structure has yet been fixed.

**Wave**: A stage of exploration in which multiple directions diverge and compete. Difference becomes visible and fluctuation appears.

**Edge**: A tense condition in which opposing elements coexist without converging into either side. Whether one can remain at this stage bears on the quality of creation.

**Vortex**: A stage in which a new coherence or order rises spontaneously out of the tension.

**Bundle**: A stage in which form is fixed and stabilizes as a reusable structure. Bundle becomes the precondition for the next Field, and the cycle continues.

The five-stage model is not limited to a particular scale, individual cognition, team activity, or social change. The investigation asks whether similar structures appear across different scales.

If these definitions already feel uncomfortable, that is itself an important point for the investigation. Holding onto that discomfort while reading can provide a more critical perspective on the results.

## 4. Results: Overall Picture

Figure 2: Psychology and Cognitive Science: What Happens When Cognition Shifts — ×5

Structural correspondence with the five-stage model was confirmed for all ten candidate theories. The strength of correspondence is as follows.

| Strength | Number of Theories | Theories |
|---|---|---|
| Strong structural correspondence | 5 | insight problem solving, Bayesian brain hypothesis, schema theory, biased competition of attention, cognitive flexibility |
| Partial correspondence | 5 | cognitive dissonance, flow experience, 4E cognition, affordances, conceptual metaphor |

Psychology and cognitive science are domains with a high degree of structural correspondence to the five-stage model. The especially notable point is that many different cognitive implementations of Edge, the tense state in which opposing elements coexist, appear together within one domain. That seems to be a feature not found in quite the same way elsewhere.

At the same time, the fact that all ten selected theories showed some correspondence may partly

reflect the possibility that the selection criteria themselves favored theories more likely to fit. This remains an interpretive caution.

## 5. Results: Findings by Theory

### 5.1 Insight Problem Solving: Tracking the Moment When Edge Is Formed

Insight problem solving is the research field that studies cases in which an initial problem representation is inappropriate, an impasse occurs, representational change takes place through relaxation of constraints or recoding, and the solution appears suddenly.

Its structural correspondence with the five-stage model is among the clearest of all ten cases.

Initial representation and problem space correspond to Field. Accumulation of error during impasse corresponds to Wave. Constraint relaxation, that is, selection of which relation to release, corresponds to Edge. Representational change and the Aha experience correspond to Vortex. Maintenance of the correct representation and consolidation as learning correspond to Bundle.

What matters here is that this correspondence is not imposed externally. It is built into the theory itself. Eye-tracking work on matchstick problems has shown that when fixation shifts from value to operator, that shift corresponds to representational change. This makes insight research a rare case in which one can objectively track what relation was newly formed at Edge.

Further, evidence that even subliminal hints can improve solution rate suggests that formation of Edge can occur without conscious awareness, as latent rewiring of relation.

**Distinctive insight from this theory**: Whereas other domains often describe Edge in terms of what meets what, insight research describes what has been released, namely constraint relaxation. Edge therefore has a double character, release and connection, and insight problem solving shows both more concretely than most theories do.

### 5.2 Cognitive Dissonance: Wave Experienced as Discomfort

Cognitive dissonance is the unpleasant psychological state produced by inconsistency among cognitive elements, and the motivation to reduce it. In the classic forced-compliance experiment, the low-reward group showed greater attitude change than the high-reward group.

In correspondence with the five-stage model, a coherent system of beliefs and attitudes corresponds to Field. Detection of inconsistency and the emergence of discomfort correspond to Wave. Exploration of reduction strategies, attitude change, behavior change, justification, or reducing importance, corresponds to Edge. Convergence on a particular interpretation and attitude change correspond to Vortex. Retention of the updated value representation corresponds to Bundle.

The distinctive contribution of this theory is that Wave is experienced most vividly as emotion, namely discomfort. In other theories, Wave is described as error or mismatch. In cognitive dissonance, subjective unpleasantness becomes the driving force.

A second distinctive point is the clarity with which multiple reduction strategies compete at Edge. One may change attitude, change behavior, justify, or downgrade importance. This branching structure shows that Edge is not a single path but a space of alternatives.

At the same time, reduction of dissonance does not always lead to attitude change, so there is a risk of reading a branching process as if it were a single line. For that reason the correspondence is partial.

### 5.3 Schema Theory: Describing Routes of Error Processing

Schema theory holds that organized structures of knowledge abstracted from past experience guide interpretation of new experience. Piaget formulated two routes of processing within this framework: assimilation, taking new information into an existing schema, and accommodation, changing the schema itself.

The correspondence with the five-stage model is clear. Existing schema correspond to Field. Mismatch, or disequilibrium, between schema and new information corresponds to Wave. The branching between assimilation and accommodation corresponds to Edge. Reorganization of schema through accommodation corresponds to Vortex. Stabilization of updated schema corresponds to Bundle.

**Distinctive insight from this theory**: Schema theory gives one of the most direct descriptions of branching in response to error. When new information does not fit the existing frame, a person either distorts the information so that it fits, assimilation, or changes the frame itself, accommodation. Edge corresponds to this branching point, and the quality of cognitive transformation depends on which route is taken.

Neisser's perceptual cycle, in which schema direct exploration and information from the environment modifies schema, can also be read directly as a circulation of Field -> Wave -> Edge.

### 5.4 The Bayesian Brain Hypothesis: The Computational Basis of Error Processing

The Bayesian brain hypothesis, or predictive processing, holds that the brain continually generates predictions from an internal model of the world and repeatedly updates so as to minimize mismatch with sensory input, prediction error. There are two routes for handling prediction error: updating the model, perceptual inference, and changing the environment through action, active inference.

In correspondence with the five-stage model, the generative model, an internal representation of the world, corresponds to Field. Prediction error corresponds to Wave. Precision weighting, the selection of which errors to attend to, corresponds to Edge. Model updating or active inference corresponds to Vortex. Stabilization of the updated world model corresponds to Bundle.

**Distinctive insight from this theory**: Precision weighting is especially important as a computational implementation of Edge. The question of which error receives weight provides a mathematical way of describing what may be happening at Edge. In that sense, the Bayesian brain hypothesis can be read as providing the computational basis of schema theory.

Its relation to schema theory is also suggestive. If one reinterprets schema as generative model, Piagetian assimilation and accommodation may be describable in a unified predictive-processing framework. That remains a hypothesis rather than a settled conclusion.

### 5.5 Flow Experience: The Subjective Experience of Vortex

Flow experience is the state of optimal experience produced by strong immersion in an enjoyable activity, marked by lowered self-consciousness, altered time sense, and a merging of action and awareness. It has been measured with instruments such as the Flow State Scale, and meta-analytic work has reported strong links between work-related flow and proactive behavior.

In correspondence with the five-stage model, the context of activity corresponds to Field. Tension between challenge and skill corresponds to Wave. Reciprocation between performance and feedback corresponds to Edge. The flow state itself corresponds to Vortex. Subsequent satisfaction and skill updating correspond to Bundle.

**Distinctive insight from this theory**: Flow is the most direct description of the subjective experience of Vortex. Especially important is that the theory specifies both the conditions under which Vortex is sustained and the conditions under which it collapses. When challenge exceeds skill too greatly, anxiety results; when it falls too far below skill, boredom results. This instability suggests that Vortex requires specific boundary conditions to maintain itself.

At the same time, if flow is treated as a discrete state, Vortex becomes the center and correspondence with the other four stages remains conceptual rather than strongly structural. For that reason the correspondence is partial.

### 5.6 Biased Competition of Attention: Selection and Inhibition Decided Together

The biased-competition model of attention holds that multiple visual stimuli compete within receptive fields, and that top-down goals and bottom-up salience bias the resolution of that competition. The representation of the selected stimulus is relatively enhanced, while nonselected stimuli are suppressed.

In correspondence with the five-stage model, the receptive field in which multiple stimuli compete corresponds to Field. Instability produced by competition corresponds to Wave. Resolution of competition through bias corresponds to Edge. Stable maintenance of the dominant stimulus representation corresponds to Vortex. Retention of selected information in working memory corresponds to Bundle. This can be read as an implementation of the five stages at the scale of milliseconds.

**Distinctive insight from this theory**: The mechanism by which attention actively suppresses what it did not choose offers an important insight. To discard is not merely to overlook passively. It is to suppress actively. Picking something back up can therefore be understood as releasing suppression and reallocating attention.

### 5.7 Cognitive Flexibility: Cutting Old Edge and Forming New Edge

Cognitive flexibility, or set shifting, is a component of executive function referring to the ability to switch between task sets. Such switching carries a cost, and residue from the previous task set can interfere with the shift.

In correspondence with the five-stage model, the current task set corresponds to Field. The demand to switch tasks corresponds to Wave. The dual process of inhibition and activation corresponds to Edge. Stable operation under the new task set corresponds to Vortex. Developmental consolidation of flexible switching ability itself corresponds to Bundle.

**Distinctive insight from this theory**: This theory describes a state in which Edge cannot be formed properly. Perseveration is a state in which one remains in an old task set and fails to form a new relation. Conversely, switching too easily corresponds to failure to hold a relation long enough. The theory therefore suggests that Edge has an optimal duration.

### 5.8 4E Cognition: Expanding the Field in Which Cognition Occurs

4E cognition is a family of positions that refuse to reduce cognition to computation inside the brain. Through the four aspects embodied, embedded, enacted, and extended, cognition is treated as a process of the whole brain-body-environment system.

In correspondence with the five-stage model, the whole brain-body-environment system corresponds to Field. The emergence of difference through interaction with the environment corresponds to Wave. Active exploration between body and environment corresponds to Edge. Constitution of cognition through action corresponds to Vortex. Stabilization of the system including the environment corresponds to Bundle.

**Distinctive insight from this theory**: 4E cognition expands Field from something inside the brain to the entire brain-body-environment system. Where other theories describe stages of process, 4E cognition describes the nature of the Field in which the process happens. Because it is a meta-theoretical framework, however, the correspondence remains largely conceptual and its precision for predicting particular processes is limited.

### 5.9 Affordances: A Field Filled with Possibilities for Action

Affordances are the possibilities for action that the environment offers an organism, defined relationally through the organism's capacities and the environment's properties. In ecological perception, perception is treated as direct pickup of invariants in the environment and does not require construction of internal representation.

In correspondence with the five-stage model, affordance space corresponds to Field. Mismatch between affordance and action capability corresponds to Wave. Exploratory contact through perception-action circulation corresponds to Edge. Discovery and use of new possibilities for action correspond to Vortex. Stabilization of fit with the environment corresponds to Bundle.

**Distinctive insight from this theory**: Gibson's theory of direct perception claims an Edge without representation. Where insight, dissonance, and schema theories treat representational change as the mechanism of Edge, Gibson bypasses representation. This contrast is a major theoretical tension within cognitive science and raises the question of how far Edge in the five-stage model depends on internal representation.

It also suggests something important about Field. Field should not be understood as empty, but as not yet differentiated while already full of possibility.

### 5.10 Conceptual Metaphor: Illuminating the Vocabulary of the Five Stages Themselves

Conceptual metaphor theory holds that abstract concepts are understood through metaphor grounded in bodily experience. Much everyday language, argument is war, time is money, is structured by mappings from bodily source domains. Image schemas are preconceptual patterns extracted from bodily experience, such as container, path, force, up-down, and they provide the basis of conceptual structure.

In correspondence with the five-stage model, accumulation of bodily experience corresponds to Field. Perception of bodily difference corresponds to Wave. Patterning through image schemas, the coupling of source and target domains, corresponds to Edge. Formation of conceptual metaphor corresponds to Vortex. Cultural stabilization as linguistically articulated abstract concept corresponds to Bundle.

**Distinctive insight from this theory**: This theory occupies a unique position within the investigation because the vocabulary of the five stages itself, Field, Wave, Edge, Vortex, and Bundle, depends on image schemas such as container, vibration, connection, rotation, and aggregation. Conceptual metaphor theory therefore functions as a mirror reflecting the linguistic basis of the five-stage model itself. Does the five-stage model describe the structure of the world, or the structure by which the world is understood through bodily metaphor? That question remains unresolved.

## 6. Cross-Domain Patterns

The following patterns recurred across multiple theories.

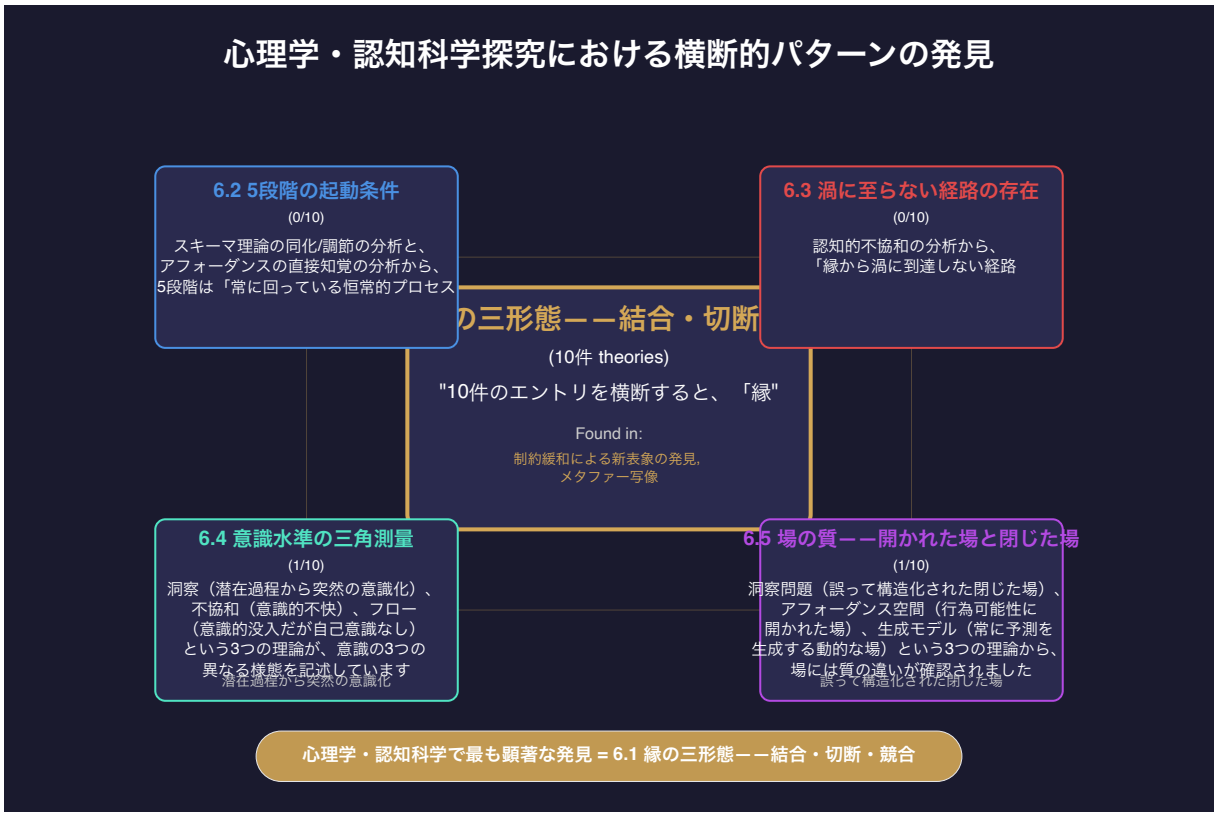


Figure 3: Psychology and Cognitive Science: What Happens When Cognition Shifts —

### 6.1 Diverse Implementations of Edge

At least six different implementations of Edge were confirmed in psychology and cognitive science.

| Implementation | Theory | What Happens at Edge |
|---|---|---|
| Constraint relaxation | Insight problem solving | Selection of which relation to release |
| Strategy branching | Cognitive dissonance | Selection among attitude change, behavior change, justification, and downgrading importance |
| Precision weighting | Bayesian brain hypothesis | Selection of which error receives weight |
| Assimilation / accommodation branching | Schema theory | Selection between fitting into the existing frame or transforming the frame |
| Competition resolution | Biased competition of attention | Simultaneous decision of selection and suppression |
| Set shifting | Cognitive flexibility | Dual process of cutting old Edge and forming new Edge |

All of these share the structure of selecting how to connect, or release, relations among elements. Yet the mechanisms of selection differ from theory to theory. This suggests that Edge is not one mechanism, but a structural feature shared by several different mechanisms.

### 6.2 Three Modes of Consciousness

Three theories together form a triangulation of different modes of consciousness.

- **Insight problem solving**: latent processing proceeds and then suddenly becomes conscious as an Aha experience
- **Cognitive dissonance**: contradiction among beliefs is experienced as conscious discomfort and motivates reduction
- **Flow experience**: consciousness remains directed toward activity in deep immersion, while self-consciousness declines

From latent processing to sudden awareness in insight, from conscious discomfort to attitude change in dissonance, and from conscious immersion to skill updating in flow, consciousness participates in different ways when cognition transforms. This suggests that consciousness may not enter the five stages uniformly.

### 6.3 Complementarity Across Levels of Description

The Bayesian brain hypothesis and schema theory capture the same phenomena at different descriptive levels. In Marr's terms, the Bayesian brain is closer to the computational and algorithmic level, while biased competition of attention is closer to the level of neural implementation. Schema theory sits between them.

The fact that structural correspondence with the five-stage model appears repeatedly across different levels of description suggests that the correspondence is not unique to one level alone. Whether the correspondences across levels are precisely isomorphic remains a question for further work.

## 7. Unresolved Questions

- **Is the five-stage model a structure of the world, or a structure of understanding?**: The question illuminated by conceptual metaphor theory, namely that the vocabulary of the five stages itself depends on image schemas, bears directly on the ontological status of the model. It may be both, but no conclusion has been reached.
- **Tension between direct perception and predictive processing**: Affordance theory, which rejects mediation by internal representation, and the Bayesian brain hypothesis, which relies on internal models, remain in tension. How far Edge depends on internal representation is part of that unresolved tension.
- **The optimal duration of Edge**: Findings from cognitive flexibility suggest that Edge has an optimal duration. Remain too long and one becomes rigid; fail to remain long enough and one becomes distractible. What determines that optimum is still unknown.
- **The boundary with neuroscience**: The Bayesian brain hypothesis at cognitive level and predictive coding at neural implementation level describe the same family of phenomena from different levels. How the boundary between domains should be drawn remains open.

## 8. Conclusion

Psychology and cognitive science are domains with high structural correspondence to the five-stage model. Correspondence was confirmed for all ten candidate theories, and in five of them the correspondence was strong.

Three main findings emerge from the investigation.

First, cognitive implementations of Edge were confirmed in six different forms: constraint relaxation, strategy branching, precision weighting, branching between assimilation and accom-

modation, competition resolution, and set shifting. These differ in mechanism but share the structure of deciding how relations are to be formed.

Second, three modes of consciousness, latent process, discomfort, and immersion, form a triangulated structure. This suggests that consciousness may participate differently at different stages of the five-stage model.

Third, structural correspondence was confirmed repeatedly at different levels of description, computational, cognitive, and neural-implementation level. This suggests that the correspondence is not tied to only one level, though the precision of the relation among levels remains to be tested.

At the same time, there are weaker areas of correspondence. 4E cognition and affordances remain mainly framework-level correspondences, with limited power to predict specific processes. The self-referential question raised by conceptual metaphor theory, namely that the vocabulary of the five stages may itself be metaphorical, remains unresolved and bears directly on the ontological status of the model.

The results of this investigation suggest that structural correspondence exists between the theories of psychology and cognitive science and the five-stage model. Whether that correspondence amounts to structural identity or remains only a similarity of description is a question for future work.